



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First/Second Semester B.Tech (CBCS) Degree Examinations

Summer Semester July 2024

MA1008-1 – CALCULUS AND LAPLACE TRANSFORMS
 (BT, CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The angle between radius vector and tangent is given by
 A) $\tan\phi = \frac{d\theta}{dr}$ B) $\tan\phi = r \frac{dr}{d\theta}$
 C) $\tan\phi = \frac{1}{r} \frac{d\theta}{dr}$ D) $\tan\phi = r \frac{d\theta}{dr}$
- The angle between the radius vector and tangent for the vector $r = ae^{\theta \cot \alpha}$ is _____
 A) $\tan \alpha$ B) $\cot \alpha$
 C) α D) θ
- The derivative of arc for the curve $y = f(x)$ is $\frac{ds}{dx} =$
 A) $\sqrt{1 - \left(\frac{dy}{dx}\right)^2}$ B) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$
 C) $1 + \left(\frac{dy}{dx}\right)^2$ D) $1 - \left(\frac{dy}{dx}\right)^2$
- The value of c got by applying Cauchy's mean value theorem for the functions $f(x) = e^x$ and $g(x) = e^{-x}$ in $[0, 1]$ is
 A) $1/2$ B) $2/3$
 C) $1/3$ D) none of these
- If $u = x^y$, then $\frac{\partial u}{\partial x}$ is
 A) 0 B) yx^{y-1}
 C) $x^y \log y$ D) $x^y \log x$
- If $f(x, y, z) = x^2 + xyz + z$ then f_x at $(1, 2, 1)$
 A) 1 B) 2
 C) 4 D) 3
- Which among the following is the definition of Jacobian of u and v w.r.t x and y?
 A) $J\left(\frac{x, y}{u, v}\right)$ B) $J\left(\frac{u, v}{x, y}\right)$
 C) $\frac{\partial(x, y)}{\partial(u, v)}$ D) $\frac{\partial(v, u)}{\partial(x, y)}$
- If $A = f_{xx}(a, b), B = f_{xy}(a, b), C = f_{yy}(a, b)$ then $f(x, y)$ will have a maximum at (a, b) if
 A) $f_x = 0, f_y = 0, AC < B^2$ and $A < 0$ B) $f_x = 0, f_y = 0, AC = B^2$ and $A > 0$
 C) $f_x = 0, f_y = 0, AC > B^2$ and $A > 0$ D) $f_x = 0, f_y = 0, AC > B^2$ and $A < 0$

9. If $L\{f(t)\} = \bar{f}(s)$, then $L\{\cos 3t\} =$ -----
 A) $\frac{3}{s^2+3^2}$ B) $\frac{s}{s^2-3^2}$
 C) $\frac{s}{s^2-3^2}$ D) $\frac{s}{s^2+3^2}$
10. $L\{t^2 e^{5t}\} =$ -----
 A) $\frac{2}{(s-5)^3}$ B) $\frac{5}{(s+1)^2}$
 C) $\frac{(2s)}{(s+5)^3}$ D) $\frac{2}{(s-1)^3}$
11. If $L\{f(t)\} = \bar{f}(s)$, then $L\{\frac{f(t)}{t}\} =$ -----
 A) $\frac{1}{s} \bar{f}(s)$ B) $\int_s^\infty \bar{f}(s) ds$
 C) $\frac{1}{s} \int_s^\infty \bar{f}(s) ds$ D) $\int_0^\infty \bar{f}(s) ds$
12. $L\{u(t+2)\} =$ -----
 A) $\frac{e^{-2s}}{s^2}$ B) e^{2s}
 C) $\frac{e^{2s}}{s}$ D) $\frac{e^{-2s}}{s}$
13. Inverse Laplace transform of $\frac{1}{(s+3)^2}$ is -----
 A) te^{-3t} B) te^{2t}
 C) e^{2t} D) e^{-2t}
14. $L^{-1}\{\frac{5e^{-3s}}{s}\} =$ -----
 A) $5t u(t-3)$ B) $5t u(t+3)$
 C) $5 u(t+3)$ D) $5 u(t-3)$
15. $L^{-1}\{\frac{(s-a)}{(s-a)^2+b^2}\} =$ -----
 A) $e^{at} \cos bt$ B) $e^{-at} \cos bt$
 C) $\frac{e^{-at} \cos bt}{b}$ D) $\frac{e^{at} \cos bt}{b}$
16. Inverse Laplace Transform of $\log\left(\frac{s+3}{s+4}\right)$ is
 A) $\frac{1}{t} [e^{-3t} - e^{-4t}]$ B) $-\frac{1}{t} [e^{-3t} - e^{-4t}]$
 C) $-\frac{1}{t} [e^{-3t} + e^{-4t}]$ D) $[e^{-3t} - e^{-4t}]$
17. One dimensional heat equation is
 A) $u_t = c^2 u_{xx}$ B) $u_{tt} = c^2 u_x$
 C) $u_t = c^2 u_x$ D) $u_{tt} = c^2 u_{xx}$
18. $x \frac{\partial z}{\partial y} + \frac{\partial^2 z}{\partial x^2} = 5 \frac{\partial^3 z}{\partial y^3}$ is a partial differential equation of order -----?
 A) 2 B) 1
 C) 4 D) 3
19. The partial differential equation formed by eliminating arbitrary constant from the equation $z = (a+x)^2 + y$ is -----
 A) $4z = p^2 + 4y$ B) $z = px - 4y$
 C) $2z = p + 4y^2$ D) $2z = p - y$

20. Solution of the partial differential equation $\frac{\partial z}{\partial x} + \cos(3x - 2y) = 0$ by direct integration is

A) $z + \frac{\sin(3x - 2y)}{3} = x f_1(y)$

B) $z - \frac{\sin(3x - 2y)}{3} = f_1(y)$

C) $z + \frac{\sin(3x - 2y)}{3} = f_1(y)$

D) $z - \frac{\sin(3x - 2y)}{3} = f_1(x)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Note to QP Setter: Each main question carries 16 Marks, maximum sub questions up to 4 levels.

Unit - I		Marks	BT*	CO*	PO*
1.	a) With the usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$	5	L*1	1	1
	b) Find the radius of curvature of the curve $r = a(1 + \cos \theta)$	5	L2	1	1
	c) Find first and second order derivatives of $z = \tan^{-1}\left(\frac{y}{x}\right)$	6	L2	2	1
2.	a) Expand the function upto 3 rd degree term $f(x, y) = \cos x \cos y$.	5	L2	2	1
	b) Using Lagrange's mean value theorem determine c for the function $f(x) = \tan^{-1} x$ in $[0, 1]$	5	L2	1	1
	c) Find the extreme values of the function $f(x, y) = x^4 + y^4 - x^2 - y^2 + 1$	6	L1	2	1
3.	a) Find the derivative of arc length of $r^2 = a^2 \cos 2\theta$	5	L1	1	1
	b) State and prove Cauchy's mean value theorem.	5	L1	1	1
	c) If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$ prove that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$	6	L2	2	1
Unit - II					
4.	a) Find $L[e^{5t} \sin 5t \cos 3t]$	5	L1	3	1
	b) If $f(t)$ is a periodic function with period T such that $f(t+T) = f(t)$ for all values of t , prove that $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$.	5	L2	3	1
	c) Find inverse Laplace Transform of $\frac{s^2+s-2}{s(s+3)(s-2)}$	6	L1	4	1
5.	a) Find the inverse Laplace transform of (i) $\log\left(\frac{s^2+1}{s(s+1)}\right)$ (ii) $-\frac{4}{s^2} e^{-s}$	5	L1	4	1
	b) Find the Laplace Transform of $\int_0^t \frac{1-\cos 2t}{t} dt$	5	L1	3	1
	c) Rewrite the following function using unit step function and find its L.T. $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4, & 2 < t < 4 \\ 0, & t > 4 \end{cases}$	6	L2	3	1
6.	a) Find Laplace Transforms of $L\{t \sin 3t \cos 2t + e^{(3t+7)}\}$	5	L1	4	1
	b) Using the convolution theorem find $L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right]$	5	L2	4	1
	c) Solve the differential equation by Laplace transform $(x''(t)) + 4x'(t) + 4x(t) = 4e^{(-2t)}; x(0) = -1, x'(0) = 4$.	6	L1	3	1

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Form a Partial differential equation by eliminating arbitrary constants from $z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ | 5 | L1 | 5 | 1 |
| b) | Solve the following higher order partial differential equation by direct integration method $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y)$ | 5 | L1 | 5 | 1 |
| c) | Derive one dimensional heat equation in the form $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ | 6 | L2 | 5 | 1 |
| 8. a) | Form partial differential equations by eliminating the arbitrary function from the equation $f(xy + z^2, x + y + z) = 0$ | 5 | L1 | 5 | 1 |
| b) | Solve the partial differential equation $\frac{\partial^2 z}{\partial y^2} - 4z = 0$ given $y = 0, z = e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$ | 5 | L1 | 5 | 1 |
| c) | Solve the following linear partial differential equation by Lagrange's method $p \cot x + q \cot y = \cot z$ | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
